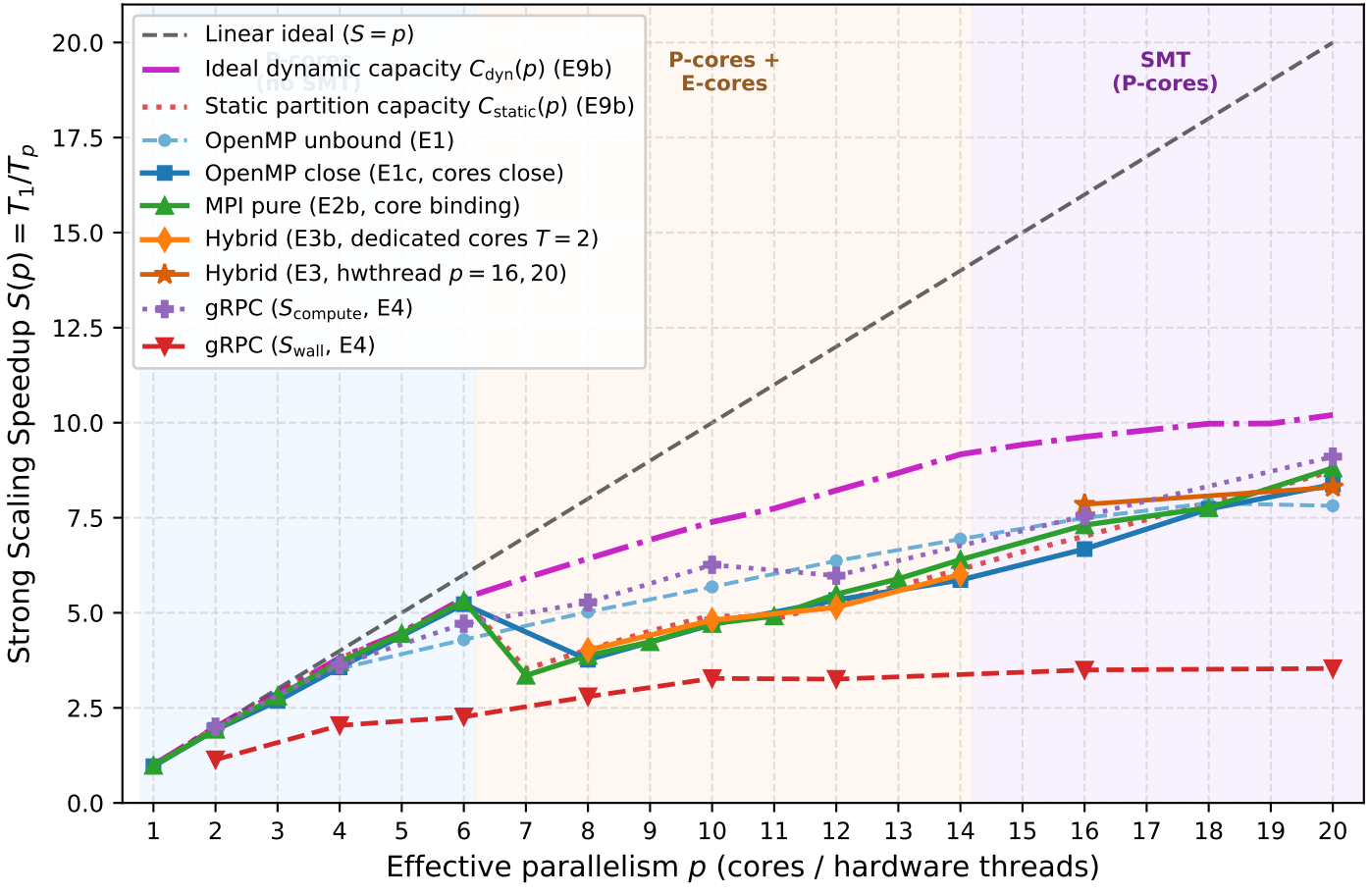
Fig. A — Comparative Strong Scaling (ECDSA P-256 Batch 20,000)
Linear ideal ($S = p$)
Ideal dynamic capacity $C_{dyn}(p)$ (E9b)
Static partition capacity $C_{static}(p)$ (E9b)
OpenMP unbound (E1)
OpenMP close (E1c, cores close)
MPI pure (E2b, core binding)
Hybrid (E3b, dedicated cores $T = 2$)
Hybrid (E3, hwthread $p = 16, 20$)
gRPC ($S_{compute}$, E4)
gRPC (S_{wall}, E4)
P-cores + E-cores
SMT (P-cores)
Strong Scaling Speedup $S(p) = T_1/T_p$
Effective parallelism p (cores / hardware threads)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20